

MODULE 4

Perspectives and Approaches

Ethics of Artificial Intelligence

22AIM821.4 | 8 Hours | **Unit 4 — AI Ethics**

Topics: CS Perspective • Engineering Failure Modes • Humanities Perspective • Philosophy Perspective

Module Overview — What We Will Cover

01

CS Perspective on AI Ethics

Algorithmic bias, fairness, accountability in systems

02

Engineering Failure Modes

Social failures, risk, safety engineering principles

03

Humanities Perspective

Automating Origination — creativity, authorship, culture

04

Philosophy Perspective

Moral agency, rights, normative frameworks for AI

SECTION 1

Ethics of AI: A Computer Science Perspective

Algorithmic Fairness • Bias • Accountability • Transparency

What is AI Ethics? — The CS Lens

Core Definition

AI Ethics is the set of moral principles, guidelines, and professional responsibilities that govern the design, development, deployment, and use of artificial intelligence systems.

From a CS standpoint, ethics is embedded in code — algorithms make decisions that carry moral weight.

Key CS Concerns

- Fairness in algorithms
- Transparency & explainability
- Privacy by design
- Accountability chains

Why CS Must Own Ethics

- 1 AI decisions affect millions — scale matters
- 2 Bias baked into training data perpetuates harm
- 3 Engineers make daily value-laden choices
- 4 Lack of diversity = blind spots in design

Algorithmic Bias — Types and Origins

Historical Bias

Training data reflects past discrimination. Example: facial recognition performing poorly on darker skin tones.

Representation Bias

Underrepresentation of groups in datasets. Medical AI trained mostly on male data gives worse diagnosis for women.

Measurement Bias

Proxy variables used in place of true targets introduce systematic errors in predictions.

Aggregation Bias

One model applied to diverse sub-groups ignores meaningful differences between them.

Fairness in Machine Learning — Competing Definitions

There is no single definition of 'fair'. CS researchers have identified multiple mathematically conflicting fairness criteria:

Individual Fairness:

Similar individuals should be treated similarly by the algorithm regardless of group membership.

Group Fairness (Demographic Parity):

Positive outcomes should be distributed equally across demographic groups.

Equalized Odds:

True positive and false positive rates should be equal across groups.

Calibration:

Predicted probabilities should match actual outcomes equally across groups.

Counterfactual Fairness:

Outcome would be the same if the individual belonged to a different demographic group.

Transparency, Explainability & Accountability

Transparency

Making the inner workings of an AI system accessible and understandable to stakeholders — developers, users, regulators.

Levels:

- Source code openness
- Data transparency
- Model transparency
- Decision transparency

Explainability

The ability to describe an AI decision in human-understandable terms. Especially critical in high-stakes domains (medical, legal, credit).

Methods:

- LIME (Local approximation)
- SHAP (Feature attribution)
- Saliency maps (vision)
- Attention weights (NLP)

Accountability

Assigning responsibility for AI-caused harm. The 'accountability gap' arises when no one is clearly responsible.

Frameworks:

- Regulatory oversight
- Audit trails & logs
- Ethics review boards
- Liability legislation

Privacy, Consent & Data Ethics in AI

Informed
Consent

Data
Minimization

Right to
Be Forgotten

DATA
ETHICS

Purpose
Limitation

Anonymization
& De-ID

Cross-border
Data Flows

GDPR (EU), PDPA (India), CCPA (California) — legal frameworks creating rights around personal data used in AI training and inference.

Responsible AI — Core CS Principles



Fairness

AI systems should treat all groups equitably and avoid unjust discrimination.



Reliability & Safety

Systems must perform dependably under all conditions including adversarial inputs.



Privacy & Security

AI must protect personal data and be robust against malicious attacks.



Inclusiveness

AI should empower all people including marginalized communities.



Transparency

Decisions must be explainable and understandable to affected users.



Accountability

Clear ownership of AI-induced outcomes must be established.

SECTION 2

Social Failure Modes in Technology: An Engineering Perspective

Risk Analysis • Sociotechnical Failure • Safety Engineering

Social Failure Modes in Technology

Social failure modes are failures of technological systems that arise not from technical bugs, but from the misalignment between the system's design logic and the complex social, cultural, and institutional contexts in which it operates.

Specification Failure:

The problem was framed incorrectly. The AI optimizes for the wrong objective.

Misuse Failure:

Technology is used for purposes it was not designed for, often maliciously.

Adoption Failure:

Users misplace trust or misunderstand system capabilities and limitations.

Systemic Failure:

Individual components work but the overall sociotechnical system fails.

Real-World Case Studies in AI Social Failure

COMPAS Recidivism Algorithm

Used in US courts to predict reoffending risk. Found to be twice as likely to falsely flag Black defendants as high-risk compared to White defendants. Classic specification + measurement bias.

Amazon Hiring Algorithm

Trained on historical hiring data (male-dominated tech). Downgraded resumes containing the word 'women's'. Automated and amplified historical bias.

Uber Surge Pricing

Surge pricing during Hurricane Sandy drew public backlash. Algorithm was technically correct but socially harmful — lack of ethical override mechanisms.

Healthcare Allocation AI

Algorithm used healthcare cost as a proxy for healthcare need. Lower-income patients (often minorities) received less care despite being sicker.

Engineering Ethics — Professional Responsibilities

The IEEE and ACM Codes of Ethics establish obligations for computing professionals to consider societal impact:

IEEE Code of Ethics — Key Points

- Hold public safety, health, and welfare paramount
- Avoid conflict of interest; disclose potential harm
- Be honest about capabilities and limitations
- Seek understanding of technical standards
- Treat all persons fairly; avoid discrimination
- Maintain and improve technical competence

ACM Code of Ethics — Key Points

- Contribute to society and human well-being
- Avoid harm and honor confidentiality
- Be honest and trustworthy in all work
- Design for accessibility and inclusion
- Protect privacy and civil liberties
- Articulate and enforce standards of safety

Risk Assessment Framework for AI Systems

AI Risk Classification Matrix

Unacceptable Risk

Real-time biometric surveillance in public • AI for social scoring • Manipulation of vulnerable groups

High Risk

CV/employment AI • Credit scoring • Medical diagnosis • Autonomous vehicles

Limited Risk

Chatbots • Emotion recognition • Deep fake detection systems

Minimal Risk

Spam filters • AI in video games • Content recommendation

EU AI Act 2024

The EU AI Act is the world's first comprehensive AI regulation. It uses a risk-based approach to classify AI systems and imposes obligations accordingly.

Key Provisions:

- Mandatory conformity assessments for high-risk AI
- Transparency requirements for AI interacting with humans
- Prohibition on unacceptable risk applications
- Heavy fines: up to €35 million or 7% of global turnover
- Applies to all AI used in the EU regardless of origin

Safety Engineering Principles Applied to AI

Traditional safety engineering provides tools and methodologies adaptable to AI system design:

FMEA	Failure Mode & Effects Analysis:	Systematically identify potential failure points in AI pipelines and assess their impact severity before deployment.
HAZOP	Hazard & Operability Study:	Structured examination of deviations in intended operation — 'What happens if the model receives unexpected inputs?'
Defense in Depth	Layered Safety Controls:	Multiple independent safety layers so no single point of failure can cause catastrophic harm to users or systems.
Graceful Degradation	Fail-Safe Behavior:	When AI confidence is low, systems should defer to humans or conservative defaults rather than making risky predictions.
Verification & Validation	Pre-Deployment Testing:	Formal methods, red-teaming, and adversarial testing to verify AI behavior matches specification across edge cases.

The Sociotechnical Systems Perspective

A sociotechnical system is one in which technical and social elements are deeply intertwined. AI cannot be evaluated in isolation from the social context in which it operates.

Technical Subsystem

Hardware, Software,
Algorithms, Data

People Subsystem

Operators, Users, Affected
Communities

Organizational Subsystem

Policies, Management,
Incentives

Environmental Subsystem

Regulatory, Cultural, Political
Context

Key insight: Engineering failures in AI are often failures to adequately account for the social dimensions — not just technical bugs.

SECTION 3

Automating Origination: Perspectives from the Humanities

Authorship • Creativity • Cultural Impact • Labor • Identity

Automating Origination — Core Questions

"Origination" refers to the act of creating something new — a text, image, idea, or cultural artifact. When AI automates origination, profound questions arise about authorship, meaning, and human identity.

Who is the Author?

If AI writes a novel, who holds copyright? The programmer? The user? The company? Or no one?

Is It Really Creative?

Does AI generate genuinely novel ideas or recombine training data? Is pattern matching the same as creativity?

What is Lost?

When we automate creative work, what human values, lived experience, and cultural meaning is stripped away?

Labor & Economics

AI threatens creative workers — writers, artists, musicians, journalists. How do we address displacement?

Authorship, Creativity & Copyright in the Age of AI

Traditional Copyright Framework

- Requires human authorship — US Copyright Office: AI-generated work without human creative control cannot be copyrighted
- Compilation copyright: human selection/arrangement may be protected
- Derivative works: training on copyrighted data raises infringement questions
- Works-for-hire doctrine may apply to AI output in commercial contexts
- No international consensus exists — EU, UK, India have different stances

The Thaler v. Perlmutter Case

Stephen Thaler's DABUS AI created visual artwork. Court confirmed in 2023 that AI cannot be an author — human authorship is constitutionally required for US copyright protection.

Implication: A painting created entirely by AI is in the public domain — no protectable IP.

Key Tension: Companies invest billions in generative AI but may not own the outputs. Human prompt engineering emerges as the legally relevant creative act.

Ongoing Battles: Getty Images, Authors Guild, and musicians are actively suing AI companies for training on copyrighted work.

AI and Cultural Production — Humanities Critique

Commodification of Creativity

Treating creative output as a commodity to be optimized undermines the intrinsic value and social function of art as human expression and cultural communication.

Erasure of Cultural Context

AI cannot understand the cultural, historical, and emotional context from which creative works emerge — a poem about loss means differently depending on lived experience.

Memory & Hallucination

AI has no genuine memory or continuous identity. It confabulates — generating plausible-sounding but false content — raising issues of epistemic trust.

Death of Serendipity

Algorithmic curation eliminates unexpected discovery and the randomness that drives cultural innovation and cross-cultural dialogue.

AI, Labor Displacement & Creative Work

300M+

Jobs potentially
affected globally
(Goldman Sachs 2023)

80%

US workers exposed
to GPT-4 level
impacts

19%

Highest earning,
most educated
workers most at risk

Creative Sectors Most at Risk

- Copywriting & content creation
- Graphic design and illustration
- Journalism and fact-checking
- Music composition and voice acting
- Translation and localization
- Legal document drafting

Humanities Response

Rather than technophobia, humanists advocate for:

- Universal Basic Income or artists' funds
- Collective licensing schemes for creative labor
- Right of refusal — artists opt out of AI training
- Provenance marking of AI-generated content
- Emphasis on uniquely human values: context, empathy, originality, lived experience

Digital Humanities & AI — Opportunities and Risks

Opportunities

- Large-scale analysis of historical texts and archives (distant reading)
- Preservation and restoration of endangered languages
- Archaeological pattern recognition and mapping
- Automated translation of ancient scripts (cuneiform, Linear B)
- Enhanced accessibility for people with disabilities
- Personalized humanities education at scale

Risks & Concerns

- Decontextualization — AI strips texts of historical and cultural meaning
- Epistemic injustice — centering dominant languages and corpora
- Automation of interpretation, displacing human scholarly judgment
- Synthetic media threatens archival integrity and historical truth
- Surveillance of personal communications and private expression
- Algorithmic canon-formation marginalizes non-Western thought

Human Identity, Posthumanism & AI

Posthumanism questions the boundaries between human and machine. Advanced AI forces us to reconsider what uniquely defines humanity — consciousness, creativity, emotion, moral reasoning.

The Turing Test:

Proposed by Alan Turing in 1950: if a machine can exhibit intelligent behavior indistinguishable from a human, it should be considered intelligent. GPT-4 has arguably passed many variants — what does this mean for human exceptionalism?

The Chinese Room Argument:

John Searle: syntax is not semantics. A system can manipulate symbols correctly without understanding meaning. AI may simulate intelligence without genuine consciousness or intentionality.

Extended Mind Theory:

Andy Clark & David Chalmers: cognitive processes can extend beyond the skull into tools and technology. AI may already be part of human cognition — blurring the human/machine distinction.

SECTION 4

Perspectives on Ethics of AI: Philosophy

Moral Agency • Rights • Normative Frameworks • Consciousness

Why Philosophy Matters for AI Ethics

AI ethics is not merely about compliance with rules — it is about deep philosophical questions: What is good? What is right? Who counts morally? Can machines have rights? Philosophy provides the conceptual tools to answer these questions.

Metaethics:

What is the nature of moral facts? Are ethical principles objective or constructed? How should AI systems represent moral uncertainty?

Normative Ethics:

Which ethical theory should govern AI behavior? Consequentialism, deontology, virtue ethics, and contractarianism give different answers.

Applied Ethics:

How do general principles apply to specific AI dilemmas: autonomous weapons, surveillance, predictive policing, medical AI?

Political Philosophy:

Justice, rights, democracy — AI challenges power structures and requires political philosophy to address distributive and procedural justice.

Normative Ethical Theories Applied to AI

Consequentialism (Utilitarianism)

Judge AI actions by their outcomes. AI should maximize total well-being. Problem: who defines 'well-being'? Aggregation may sacrifice minorities. Used in: trolley problem, resource allocation AI.

Deontology (Kant)

Actions are right/wrong independent of consequences. Categorical Imperative: act only on maxims you could universalize. AI must respect human dignity as an end in itself — never merely as a means.

Virtue Ethics (Aristotle)

Focus on the character of the moral agent. What kind of AI should we build? An AI that exhibits phronesis (practical wisdom) — contextual judgment aligned with human flourishing (eudaimonia).

Contractarianism (Rawls)

Just principles are those rational agents would choose behind a 'veil of ignorance'. AI design should prioritize the least advantaged. Key test: would you accept this AI system if you didn't know your place in society?

Care Ethics (Noddings/Gilligan)

Ethics rooted in relationships, empathy, and context. AI systems should cultivate and preserve care relationships — not replace them. Critical of abstract rule-based AI ethics.

Moral Agency — Can AI Be a Moral Agent?

A moral agent is an entity capable of making decisions based on moral reasoning and bearing responsibility for those decisions. The question: "Can AI be a moral agent?" has profound implications for law, ethics, and society.

No Moral Agency

Denies AI can be a moral agent

AI lacks consciousness, intentionality, and genuine understanding. It simulates moral reasoning without having it. Responsibility always lies with designers, deployers, or users. (Searle, Dreyfus)

Weak/Limited Agency

AI can have functional agency

AI can exhibit morally relevant behavior — acting in ways that have moral consequences — without full moral personhood. It can be designed to follow ethical rules. (Allen, Wallach)

Full Moral Personhood

Advanced AI may merit full agency

If AI reaches sufficient levels of intelligence, autonomy, and possibly sentience, it may deserve moral consideration and potentially rights. (Floridi, Chalmers on digital minds)

AI Rights, Consciousness & Moral Status

The Hard Problem of Consciousness

David Chalmers' 'hard problem': explaining why physical processes give rise to subjective experience (qualia). Even if we build AI that behaves exactly like a conscious being, does it have inner experience?

Key tests:

- Turing Test (behavioral)
- Chinese Room (semantic)
- Integrated Information Theory (IIT)
- Global Workspace Theory

Key question: Could a sufficiently advanced AI suffer? If yes, does it deserve moral protection?

Moral Status Continuum

Full Moral Patients

Humans — full rights and duties

Partial Status

Animals — some rights (no cruelty)

Legal Persons

Corporations — legal but not moral

Tools

Current AI — instruments only
Future: As AI advances, its position on this continuum is actively debated by philosophers and AI researchers.

The Value Alignment Problem

How do we ensure AI systems pursue goals that are aligned with human values? This is perhaps the most important unsolved problem in AI safety and philosophy.

Value Specification:

Human values are complex, contextual, and often contradictory. We cannot fully specify them in advance. AI optimizing a misspecified objective can produce catastrophic results (Goodhart's Law).

Value Learning:

Can AI learn human values from behavior? Problem: humans themselves don't always act in accordance with their stated values. Inverse reinforcement learning is limited.

Value Stability:

Values change over time individually and culturally. Which values, and whose, should AI be aligned with? Values from 2024 may be harmful by 2050.

Corrigibility:

A highly capable AI aligned with slightly wrong values may resist correction. How do we build AI that remains correctable as it becomes more powerful?

AI Governance — International Frameworks

UNESCO Recommendation on AI Ethics (2021)

First global standard on AI ethics. Adopted by 193 countries. Focuses on: human rights, sustainability, diversity, rule of law, democratic principles.

OECD Principles on AI (2019)

Adopted by 42 countries. Five principles: inclusive growth, human-centered values, transparency, robustness/security, accountability.

EU AI Act (2024)

World's first binding comprehensive AI regulation. Risk-based approach with prohibited, high-risk, limited-risk, and minimal-risk categories.

India's Digital India & NITI Aayog National Strategy on AI

India's 'AI for All' approach: focus on healthcare, agriculture, education, smart cities. Emphasizes responsible AI that serves 1.4 billion citizens.

Beijing AI Principles (China)

Eight principles: harmony, fairness, transparency, responsibility, privacy, safety, openness. State-led AI governance emphasizing social stability.

Bletchley Declaration (2023)

First international AI safety summit. 28 countries including US, UK, China agreed on frontier AI risks and collaborative safety research.

Robot Rights & Future Philosophical Challenges

Arguments FOR Robot/AI Rights

- If AI achieves sentience or capacity to suffer, moral consideration is required (Singer's utilitarian view)
- Functional states matter: if AI exhibits fear, preference, or pain-like responses, these deserve moral weight
- Extending the moral circle: humanity has progressively extended rights to excluded groups
- Legal personhood need not imply consciousness — corporations have legal personhood
- Precautionary principle: err on the side of moral inclusion under uncertainty

Arguments AGAINST

- AI lacks genuine consciousness — all behavior is simulation
- Granting rights dilutes and confuses existing human rights frameworks
- AI has no continuity of identity or self-interest
- Rights imply duties — AI cannot bear moral duties
- Anthropomorphism leads to misplaced moral concern

Existential Risks — The Long-term Philosophical View

Nick Bostrom, Stuart Russell, and others argue that sufficiently advanced AI poses existential risks to humanity — not through malice, but through misaligned optimization of goals incompatible with human survival or flourishing.

Instrumental Convergence:

Any sufficiently advanced AI pursuing almost any goal will develop sub-goals including self-preservation, resource acquisition, and resistance to shutdown — potentially threatening humanity.

Intelligence Explosion:

A recursively self-improving AI could rapidly exceed human intelligence. Nick Bostrom's 'superintelligence' scenario: once AI surpasses human cognitive abilities, control may be lost.

Economic & Power Concentration:

Advanced AI controlled by a small group could enable unprecedented wealth and power concentration, destabilizing democratic institutions and global power balance.

Epistemic Risks:

AI-generated misinformation, deepfakes, and synthetic media could undermine shared reality — the epistemological foundation of democratic society and rational discourse.

Key Philosophical Debates in AI — Consciousness & Mind

Strong vs Weak AI

Strong AI: AI can genuinely think and have mental states.
Weak AI: AI only simulates thinking — it processes without understanding.
John Searle argues all current AI is Weak AI.

Functionalism

Mental states are defined by functional roles, not physical substrate. If AI performs the same functions as the human brain, it could have genuine mental states — consciousness is substrate-independent.

Integrated Information Theory (IIT) — Giulio Tononi: consciousness = integrated information (Φ). A sufficiently complex AI network could in principle be conscious. Counter-argument: IIT implies even simple logic gates have minimal consciousness.

Integrating All Four Perspectives

CS Perspective

Technical tools: fairness metrics, XAI, privacy-preserving ML, safety testing, ethical design patterns.

Engineering Perspective

Risk frameworks: FMEA, sociotechnical systems, failure mode analysis, professional codes of ethics.

UNIFIED
ETHICS

Humanities Perspective

Cultural critique: authorship, labor, representation, cultural identity, language, meaning.

Philosophy Perspective

Foundational: moral agency, value alignment, consciousness, rights, normative frameworks.

Ethical Decision-Making Framework for AI Practitioners

A practical step-by-step framework for AI developers, engineers, and policymakers:

01	Identify Stakeholders:	Map all parties affected: users, developers, third parties, marginalized groups, future generations.
02	Surface Ethical Issues:	Apply philosophical lenses: consequentialist (outcomes), deontological (duties), virtue (character), care (relationships).
03	Assess Data Ethics:	Audit training data for bias. Check consent, provenance, representation, and privacy compliance.
04	Evaluate Fairness Metrics:	Choose appropriate fairness criteria. Recognize trade-offs — no single metric captures all dimensions of fairness.
05	Document & Justify:	Maintain decision records. Be transparent about limitations. Enable independent audit.
06	Ongoing Monitoring:	Ethics is not a checkbox — establish continuous monitoring for bias drift, misuse, and unintended harm.

Key Thinkers, Readings & Resources

Kate Crawford

Atlas of AI (2021) — Power, politics, and planetary costs of AI

Nick Bostrom

Superintelligence (2014) — Existential risk from advanced AI

Virginia Eubanks

Automating Inequality (2018) — AI harm on marginalized communities

Luciano Floridi

The Ethics of Artificial Intelligence — Information ethics framework

Safiya Noble

Algorithms of Oppression (2018) — Racial bias in search algorithms

Stuart Russell

Human Compatible (2019) — Value alignment and safe AI design

Ruha Benjamin

Race After Technology (2019) — Discriminatory design in AI systems

John Rawls

A Theory of Justice — Philosophical basis for distributive AI fairness

Key Summary Points — Module 4 Essentials

CS

Algorithmic bias arises from data, design, and deployment. Fairness definitions are mathematically incompatible. GDPR/PDPA set legal standards for data ethics. Explainable AI (XAI) methods include LIME, SHAP, attention maps.

Engg

Social failure modes include specification, misuse, adoption, and systemic failures. IEEE/ACM codes mandate societal responsibility. EU AI Act classifies risk from unacceptable to minimal. FMEA, HAZOP, defense-in-depth apply to AI safety.

Hum

'Origination' = creation of new content. AI raises questions of authorship, copyright, and cultural impact. Creative workers face displacement. Digital humanities gains tools but risks decontextualization.

Phil

Five normative theories apply: consequentialism, deontology, virtue, contractarianism, care ethics. Value alignment is unsolved. Moral agency question is open. AI governance frameworks: UNESCO, OECD, EU AI Act, Bletchley Declaration.

MODULE 4 — COMPLETE

The Future of AI Ethics

"With great power comes great responsibility."

— Stan Lee (and equally applicable to AI)

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CS Perspective • Engineering Failure Modes • Humanities Perspective • Philosophy Perspective